



Chapter 6: Geometrical figures

Drill practice and mixed revision of all 10 lessons · 82
questions

NSW syllabus outcome

Stage 4 Measurement and space: MA4-ANG-C-01 (angle
relationships, triangles and quadrilaterals) and transformations
of shapes on the Cartesian plane and grids.

NAME

CLASS



Homework. Part 1 is **drill practice**: quick questions on the main skill of each lesson in Chapter 6. Write each answer on its line. Part 2 is **mixed revision**, from **Easy** to **Challenging**. Each question is tagged with its lesson, so you can look back at that lesson's summary if you get stuck.

YOU DO On your own: Part 1, drill practice.

QUESTIONS 1–30

6.01 Transforming shapes**Easy** Translate (slide) the point. Write where it ends up.**1** $(-5, 0)$: 1 right, 2 up**2** $(1, 0)$: 4 left**3** $(-1, 4)$: 6 left, 2 up**Medium** Rotate (turn) the point about the origin.**4** $(5, -2)$, 90° clockwise**5** $(4, -1)$, 180° **6** $(-2, 4)$, 90° clockwise**6.02 Combined transformations****Easy** Name the transformation: translation, reflection or rotation.**7** a lift going up**8** a mirror image**Medium** Do both transformations in order. Where does the point end up?**9** $(4, 1)$: 1 right, then reflect in the x-axis**10** $(4, -1)$: 3 right, then reflect in the x-axis**6.03 Line symmetry****Easy** How many axes (lines) of symmetry?**11** regular octagon**12** semicircle**13** isosceles trapezium**Medium** How many axes of symmetry does a regular polygon with this many sides have?**14** 5 sides**15** 13 sides**16** 4 sides**17** 15 sides**6.04 Rotational symmetry****Easy** Order of rotational symmetry (write 1 for none).**18** kite**19** scalene triangle**20** square**Medium** A regular polygon has this many sides. Write its order of rotational symmetry and the smallest angle of turn.**21** 5 sides**22** 9 sides**23** 24 sides**6.05 Classifying triangles****Easy** Classify by angles: acute, right or obtuse.**24** 110° , 41° , 29° **25** 81° , 34° , 65° **26** 65° , 81° , 34° **27** 29° , 129° , 22° **Medium** Classify by sides and by angles.**28** angles 60° , 60° , 60° **29** angles 90° , 45° , 45° **30** angles 130° , 31° , 19°

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 31–57

6.06 Angle sum of a triangle**Easy** Two angles of a triangle are given. Find the third. (Angles add to 180° .)**31** 58° and 66° **32** 69° and 76° **33** 52° and 113° **34** 58° and 62° **Medium** Isosceles triangle: find the missing angles.**35** base angle 28° **36** base angle 41° **6.07 Exterior angle of a triangle****Easy** The two interior opposite angles are given. Find the exterior angle (add them).**37** 61° and 61° **38** 82° and 16° **39** 58° and 28° **40** 55° and 100° **Medium** The exterior angle and one interior opposite angle are given. Find the other interior opposite angle.**41** exterior 74° , 23° **42** exterior 170° , 102° **6.08 Quadrilaterals****Easy** Name the quadrilateral.**43** diagonals cross at 90° and one diagonal bisects the other**44** opposite angles equal, no right angles, sides not all equal**Medium** How many pairs of parallel sides? How many right angles?**45** parallelogram**46** isosceles trapezium**47** square**6.09 Angle sum of a quadrilateral****Easy** Three angles of a quadrilateral are given. Find the fourth. (Angles add to 360° .)**48** 87° , 106° , 96° **49** 76° , 120° , 111° **50** 63° , 75° , 123° **Medium** A kite has two equal opposite angles, x . Find x .**51** other angles 123° and 67° **52** other angles 74° and 54° **53** other angles 62° and 80° **6.10 Properties of quadrilaterals****Easy** True or false?**54** Co-interior angles in a parallelogram add to 180° .**55** A kite has two pairs of equal opposite sides.**Medium** Use the properties to find the missing value.**56** square: perimeter 48 cm. The side?**57** isosceles trapezium: one base angle is 65° . The other base angle?

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 58–67

58 **Easy**

A point at (3, 2) is translated. Where is it now?
2 left, 4 up Lesson 6.01

59 **Easy**

Give the order of rotational symmetry.
rhombus Lesson 6.04

60 **Easy**

The interior opposite angles are given. Find the exterior angle.
47° and 68° Lesson 6.07

61 **Easy**

True or false?
A rectangle is a parallelogram. Lesson 6.10

62 **Easy**

How many axes of symmetry?
rhombus Lesson 6.03

63 **Easy**

Two angles of a triangle are given. Find the third.
38° and 97° Lesson 6.06

64 **Easy**

Three angles of a quadrilateral are given. Find the fourth.
110°, 75°, 105° Lesson 6.09

65 **Easy**

Name the transformation.
a slide 4 units up Lesson 6.02

66 **Easy**

Classify the triangle by its angles.
25°, 35°, 120° Lesson 6.05

67 **Easy**

Name the quadrilateral.
a rectangle with 4 equal sides Lesson 6.08

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 68–77

- 68** **Medium** A point is at (2, 3) on a grid. Where is it after a translation 4 right and 1 down? Lesson 6.01

- 70** **Medium** The exterior angle is 125° and one interior opposite angle is 60° . Find the other. Lesson 6.07

- 72** **Medium** How many axes of symmetry does a kite have? Lesson 6.03

- 74** **Medium** Three angles are 100° , 85° and 95° . Find the fourth. Lesson 6.09

- 76** **Medium** Classify a triangle with angles 90° , 45° and 45° . Lesson 6.05

- 69** **Medium** Does a kite have rotational symmetry? Lesson 6.04

- 71** **Medium** State one property of the diagonals of a rhombus. Lesson 6.10

- 73** **Medium** A right-angled triangle has an angle of 28° . Find the other angle. Lesson 6.06

- 75** **Medium** A point (1, 2) is translated 3 right then 4 up. Where is it? Lesson 6.02

- 77** **Medium** Name a quadrilateral with exactly one pair of parallel sides. Lesson 6.08

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 78–82

- 78** **Challenging** Describe what a 90° clockwise rotation does to an arrow pointing up. Lesson 6.01

- 80** **Challenging** How many axes of symmetry does a regular hexagon have? Lesson 6.03

- 82** **Challenging** Classify a triangle with angles 110° , 35° and 35° by sides and angles. Lesson 6.05

- 79** **Challenging** The exterior angle is 140° . Find the adjacent interior angle. Lesson 6.07

- 81** **Challenging** A parallelogram has an angle of 65° . Find the other angles. Lesson 6.09